

A ROBUST ESTIMATOR FOR SPREAD

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SUMMARY

Estimators are proposed for measures of spread of the type introduced in Lemmer (1977), and their properties are examined. It is also shown that confidence intervals for these measures are much more robust than confidence intervals for the population standard deviation σ when the population is unknown.

1. Introduction and main results

For a random sample $X_1, \dots, X_n$ from a distribution with unknown continuous c.d.f. F the number A defined by

$$P[(X_i - X_j)^2 > A^2] \equiv P[|X_i - X_j| > A] = 0,1 \text{ for } i \neq j$$

has been proposed in Lemmer (1977) as a measure of dispersion or spread. (A is a measure of spread in the sense of Bickel and Lehmann (1975)). The test for the null hypothesis $H_0: A = A_0$ was based on the number of absolute differences $|X_i - X_j|$, $1 \leq i < j \leq n$, exceeding A_0 .

We now consider the class of measures $\{A_q, 0 < q < 1\}$ with A_q defined by $P(|X_i - X_j| > A_q) = 1 - q$ and base our estimators and confidence intervals for A_q on the absolute differences $|X_i - X_j|$, $1 \leq i < j \leq n$. Denote these $N = \binom{n}{2}$ differences by $Z_1, \dots, Z_N$. Obviously these random variables are identically (but not all independently) distributed with common c.d.f. denoted by F_Z . Clearly $F_Z(A_q) = q$, so A_q can be estimated by

$$Q_q = \begin{cases} Z_{(Nq)} & \text{if } Nq \text{ is an integer} \\ Z_{([Nq + 1])} & \text{otherwise,} \end{cases}$$

where $Z_{(1)}, \dots, Z_{(N)}$ are the ordered Z_i , $i = 1, \dots, N$ and $[x]$ indicates the largest integer $\leq x$.

We define the relative efficiency of Q_q with respect to the sample standard deviation

tion S as the inverse ratio of their standardized mean squared errors:

$$[n E(S - \sigma)^2 / \sigma^2] / [n E(Q_q - A_q)^2 / A_q^2].$$

For the construction of confidence intervals, consider $U'_n = \sum_{i < j} I(|X_i - X_j| > A_q)$ with $I(\cdot)$ the indicator function.

Let ℓ_α be the largest integer such that $P(U'_n \leq \ell_\alpha) \leq \alpha$ and u_α the smallest integer for which $P(U'_n \geq u_\alpha) \leq \alpha$. Denote the ordered $|X_i - X_j|$ for $1 \leq i < j \leq n$ by

$$Z_{(1)} < \dots < Z_{(N)}.$$

$$\text{Now } U'_n \geq \ell_\alpha \Rightarrow Z_{(N+1-\ell_\alpha)} > A_q$$

$$\text{and } U'_n \leq u_\alpha \Rightarrow Z_{(N-u_\alpha)} < A_q$$

$$\text{so that } P(\ell_\alpha \leq U'_n \leq u_\alpha) = P(Z_{(N-u_\alpha)} < A_q < Z_{(N+1-\ell_\alpha)}) = 1 - 2\alpha.$$

Since the distribution of U'_n is unknown, ℓ_α and u_α cannot be determined directly. In Lemmer (1977) it has been proven that $U = (U_n - p) / V$ is asymptotically $n(0,1)$ distributed where $U_n = U'_n / N$, $p = 1 - q$,

$V^2 = [2p + 2(3-2n)p^2 + 4(n-2)I_{20}] / n(n-1)$ and with I_{20} to be determined as explained in the paper mentioned. Thus for large n we have approximately, with $z_\alpha = \Phi(1-\alpha)$:

$$U \leq z_\alpha \Rightarrow U'_n \leq N(p + z_\alpha V)$$

$$\text{and } U \geq -z_\alpha \Rightarrow U'_n \geq N(p - z_\alpha V).$$

Using $\langle x \rangle$ to denote the smallest integer $\geq x$, we have

$$\begin{aligned} P(-z_\alpha \leq U \leq z_\alpha) &= P(N(p - z_\alpha V) \leq U'_n \leq N(p + z_\alpha V)) \\ &\doteq P([N(p - z_\alpha V)] \leq U'_n \leq \langle N(p + z_\alpha V) \rangle) \\ &= P(Z_{(N - \langle N(p + z_\alpha V) \rangle)} < A_q < Z_{(N+1 - [N(p - z_\alpha V)])}) \end{aligned}$$

so that an approximate $(1 - 2\alpha)$ confidence interval for A_q is given by

$$(Z_{([N(q-z_{\alpha}V)])}) \cdot Z_{([N(q+z_{\alpha}V)+1])}.$$

Naturally, if $N(q+z_{\alpha}V)+1 > N$ the upper boundary is to be taken as $Z_{(N)}$.

2. Monte Carlo Study

Due to the complexity of the distribution of S for most distributions the relative efficiency of Q_q with respect to S cannot be calculated exactly, but has to be estimated by simulation. For every distribution F and every value of q , m samples of size n are generated and the values Q_{qj} and S_j , $j = 1, \dots, m$ calculated. The relative efficiency of Q_q with respect to S is estimated by

$$[\frac{n}{m} \sum_{j=1}^m (S_j - \sigma)^2 / \sigma^2] / [\frac{n}{m} \sum_{j=1}^m (Q_{qj} - A_q)^2 / A_q^2].$$

In order to calculate the value of A_q we use the relationship between f and f_Z given in table 1.

TABLE 1

f	f_Z
Uniform $[0,1]$	$f_Z(z) = 2 - 2z, \quad 0 \leq z \leq 1$
Normal $(\mu, \sigma^2/2)$	$\frac{1}{2}$ normal (σ^2)
Exponential (a)	Exponential (a) .

For f the $\chi^2(8)$ distribution the values of A_q have been calculated by numerical integration.

The estimated relative efficiencies are given in table 2 for the normal, uniform, $\chi^2(8)$ and exponential distributions.

Each Monte Carlo calculation consisted of $m = 1\,000$ repetitions on a Univac 90/60 computer using the RANDU or GAUSS subroutines. In the case of the normal or uniform distribution $q = 0,90$ would obviously be the best choice, but for the $\chi^2(8)$ and exponential distributions q should be smaller. If F is unknown a choice of $q = 0,80$ would probably be the best overall choice. By using graphic methods to get the differences $X_i - X_j$ it is fairly easy to calculate Q_q .

In practice the construction of a confidence interval is generally much more useful than the calculation of an estimate. There are two important criteria when working with confidence intervals, namely their true confidence coefficients and the lengths of the intervals.

TABLE 2
Estimated relative efficiency of Q_q with respect to S

n	q	Distribution			
		Normal	Uniform	$\chi^2(8)$	Exponential
10	0,90	0,96	0,96	0,69	0,63
	0,85	0,89	0,85	0,62	0,50
	0,80	0,87	0,72	0,90	0,84
	0,75	0,79	0,64	0,84	0,70
	0,70	0,76	0,57	0,79	0,64
	0,65			0,76	0,60
	0,60			0,87	0,78
	0,55			0,84	0,75
24	0,90	0,95	0,88	0,77	0,77
	0,85	0,92	0,74	0,86	0,84
	0,80	0,89	0,66	0,95	0,92
	0,75	0,86	0,61	1,03	0,96
	0,70	0,82	0,57	1,03	0,96
	0,65			1,06	1,02
	0,60			1,09	1,03
	0,55			1,08	1,06

We now consider the practical situation where F is unknown. Many a statistician would nevertheless work with σ and its confidence interval based on the normality assumption. In table 3 we have the empirical confidence coefficients of nominal 95% confidence intervals for σ and the A_q 's.

We notice that the confidence intervals for A_q are much more robust (the confidence coefficients are closer to 0,95) than those for σ , especially in the case of skewed distributions.

We now define L_σ , the standardized length of a confidence interval for σ , as the length of the confidence interval divided by σ , and similarly LA_q for A_q . We may use the ratio $R1 = LA_q/L_\sigma$ as criterion to judge how the length of the interval for A_q compares with that for σ . If we calculate $R1$ in the case of the $\chi^2(8)$ distribution with $n = 24$ and $q = 0,8$ we find a relative length of 1,41. However, $R1$ does not give a valid comparison, because a 92,2% confidence interval for $A_{0,8}$ is compared with a 87,1% confidence inter-

val for σ (both are nominal 95% intervals). By means of a trial and error procedure (by guessing α) we obtain also a 92,2% confidence interval for σ and calculate R2, the ratio of LA_q with respect to the standardized length of this interval, and obtain $R2 = 1,2$. Similar ratios R2 have been calculated for all cases under consideration and are presented in table 3.

TABLE 3
Empirical confidence coefficients of 95% confidence
intervals and relative lengths R2

n	q	Distribution							
		Normal		Uniform		$\chi^2(8)$		Exponential	
		E.c.c.	R2	E.c.c.	R2	E.c.c.	R2	E.c.c.	R2
10	0,90	0,790	1,0	0,825	1,0	0,708	1,1	0,538	0,9
	0,85	0,874	1,0	0,906	1,0	0,838	1,1	0,697	1,1
	0,80	0,930	1,1	0,963	1,0	0,917	1,0	0,823	1,0
	0,75	0,932	1,1	0,965	1,1	0,926	1,0	0,877	1,0
	0,70	0,914	1,1	0,959	1,1	0,908	1,1	0,874	1,2
	0,65					0,887	1,2	0,874	1,2
	0,60					0,947	1,1	0,927	1,2
	0,55					0,930	1,1	0,914	1,2
	For σ :	0,951		0,995		0,910		0,775	
24	0,90	0,930	1,3	0,955	1,1	0,893	1,2	0,860	1,2
	0,85	0,933	1,2	0,954	1,1	0,893	1,3	0,906	1,4
	0,80	0,937	1,1	0,953	1,1	0,922	1,2	0,921	1,4
	0,75	0,943	1,1	0,960	1,2	0,922	1,1	0,923	1,2
	0,70	0,936	1,1	0,957	1,2	0,924	1,0	0,933	1,1
	0,65					0,917	1,0	0,928	1,0
	0,60					0,916	1,0	0,932	1,0
	0,55					0,915	1,0	0,928	1,0
	For σ :	0,959		0,995		0,871		0,752	

A value 1,2 means that the standardized length of the confidence interval for A_q is 20% longer than that for σ . Thus if we knew the distribution we could use that value of α that would give a confidence interval for σ with the desired confidence coefficient.

Under these circumstances the confidence intervals for A_q are generally longer than the comparable confidence intervals for σ . But in practice we do not know the distribution, so we do not have the latter (92,2%) confidence interval for σ , but only the former (87,1%) confidence interval. Thus the R2 values are valid for comparison purposes, but they are only of academic interest.

Regarding the choice of q there does not seem to be any clear indication of a specific value that would be the best. Since the empirical confidence coefficients are generally not satisfactory for q very large, we again suggest using $q = 0,8$ as in the estimation case above. From the results presented the reader can judge the merits of using a confidence interval for $A_{0,8}$ relative to using a confidence interval for σ .

Other methods of confidence interval construction have also been examined, including the following: Divide the sample $X_1, \dots, X_n$ randomly into $v = [n/2]$ pairs and denote the absolute difference within pairs by $Y_1, \dots, Y_v$, which are then independant random variables with c.d.f. F_Z . For given n, q, r and $s (> r)$ the confidence coefficient of the interval $(Y_{(r)}, Y_{(s)})$ is given by

$$P(Y_{(r)} < A_q < Y_{(s)}) = \sum_{i=r}^{s-1} \binom{v}{i} q^i (1-q)^{v-i}.$$

The construction of confidence intervals by this method is very easy and quick and give intervals with remarkably accurate confidence coefficients. Unfortunately these intervals are too long compared to those presented in the paper and are therefore not recommended for general use.

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